

Name: _____

Collaborators: _____

THAYER SCHOOL OF ENGINEERING • DARTMOUTH

ENGS 28, W26

LAB 8 — SERVO MOTORS

Due: Tuesday March 3, 2026 2:15 PM

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1 Overview

Servomotors are used to achieve precise position control in mechanical and other systems. In this lab you will use a servo to point two photosensors at a light source, and track the motion of the source.

1.1 Place in course

You have now used PWM to control the speed of a brushed DC motor. This lab will introduce a new type of motor and will demonstrate another use of PWM.

1.2 Learning Objectives

- Develop familiarity with the advantages and limitations of servomotors.
- Build intuition about the operation of a simple feedback system.

1.3 Equipment

- Nucleo and breadboard
- TowerPro SG92R microservo and solar tracker fixture
- Two photocells (light-dependent resistors)
- Two $10K\Omega$ resistors
- 9V power supply
- Alligator-to-breadboard pin jumpers
- A small flashlight (the one on your phone will work)

2 Technical Study - Photocells

Photocells (also known as a CdS photocell or a Light Dependent Resistor (LDR)), are resistors whose resistance is related to the amount of light striking the Cadmium Sulfide (CdS) surface. These sensors are good at detecting a relative light level (or the presence or absence of light), but if you wanted to make a precise measurement of lux, you would need a more sophisticated sensor. For the purposes of this lab however, the photocells in your kit will work well.

In this technical study, you will design a simple light-to-voltage sensor using a photocell in a voltage divider, as seen in Figure 1

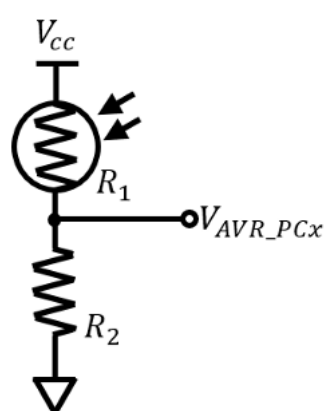


Figure 1: Photocell Voltage Divider

Record your answers to the following questions in the space provided.

Deliverable 1: Datasheet Min Max (1 pt)

Using the datasheet, record the minimum dark resistance and minimum illuminated resistance for R_1 .
 Dark: _____ Light: _____

Deliverable 2: Measured Resistance (1 pt)

Using your multimeter and a flashlight, record the dark resistance and illuminated resistance for R_1 .
 Dark: _____ Light: _____

Deliverable 3: Discrepancies (1 pt)

Why might the measured values disagree with the datasheet?

Deliverable 4: Calculated R (1 pt)

On page 133 of Williams, it is recommended that the value of R_2 be set as the geometric mean of R_{dark} and $R_{illuminated}$, i.e. $R_2 = \sqrt{R_{dark} \times R_{illuminated}}$. Based on the values you measured, compute R_2 . Answer:

Deliverable 5: Expected Voltages (1 pt)

Recall that the equation for the voltage divider as drawn above is $\frac{V_1}{V_{cc}} = \frac{R_2}{R_1 + R_2}$. Using the values you measured, roughly what are the voltages that you would expect to see at V_1 when the sensor is illuminated and when the sensor is dark, assuming $V_{cc} = 3.3V$? Dark: _____ Light:

Deliverable 6: Actual Voltages (1 pt)

You are limited by the discrete resistor values made available for your use in the laboratory, so you might not have the exact value you calculated. Assume that you have a $10k\Omega$ resistor available to you. Repeat the calculations from the previous deliverable. Is using a $10k\Omega$ going to be a problem?

Dark: _____ Light: _____

Is this a problem?

Wire the two photosensors and $10K\Omega$ resistors on your breadboard as shown in Figure 2.

Write a program to measure the two sensor voltages and print them on your screen. This should be an easy two-channel modification of the earlier A/D + potentiometer program. A one-second sampling period is fine.

Wire the sensor voltages to analog channels 1 and 2 of the AD2, to display the voltages with Waveform's voltmeter function.

Deliverable 7: Measured Values (1 pt)

Do your measured values for each channel fall within the expected range you calculated earlier?

Record your observed voltages here. Sensor 1: _____ Sensor 2:

Slowly sweep a flashlight beam (the one on your phone is fine) across the two sensors and observe how the voltages change. Note the maximum and minimum voltages. How do they compare to your theoretical maximum and minimum voltages?

Deliverable 8: Measured Min and Max (1 pt)

Record your observed voltages here. Sensor 1: _____ Sensor 2:

3 Technical Study - Servo Motors

This section repeats the class activity from Day 15. There is nothing to turn in, just make sure that everything works as intended. The complete setup is shown on the next page.

Refer to Figure 4 for the following steps.

1. Modify your program from Day 15 to generate a PWM signal due to a changing potentiometer input on analog channel 3. The PWM signal must meet the following timing specifications shown in Figure 3. Verify that the PWM signal is correct with Waveforms before moving on.

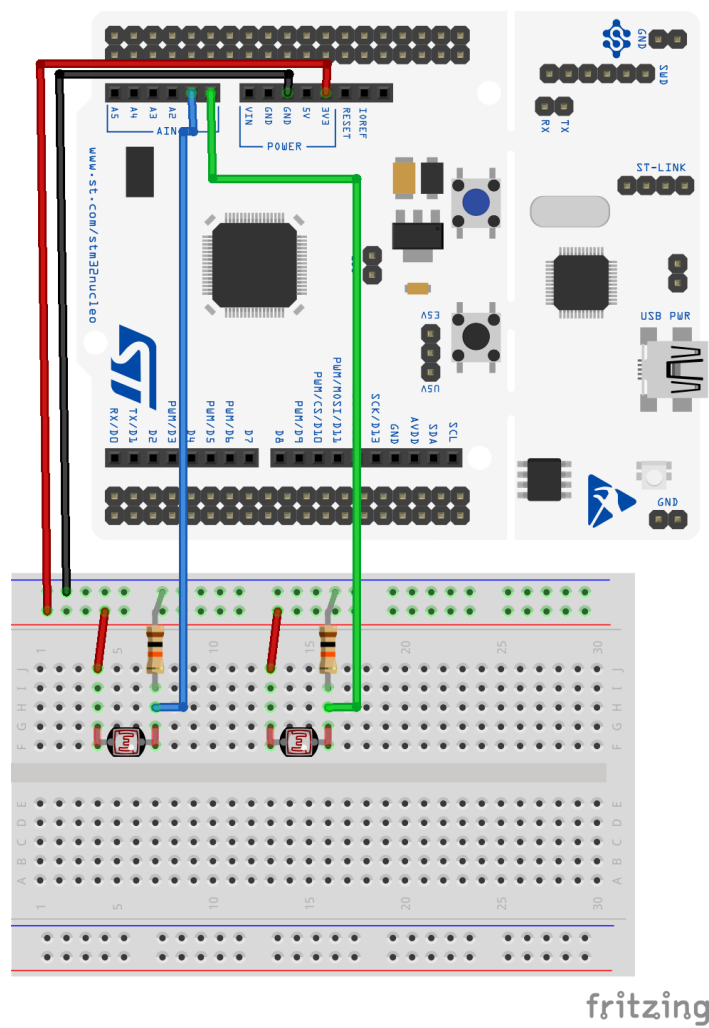


Figure 2: Test Circuit

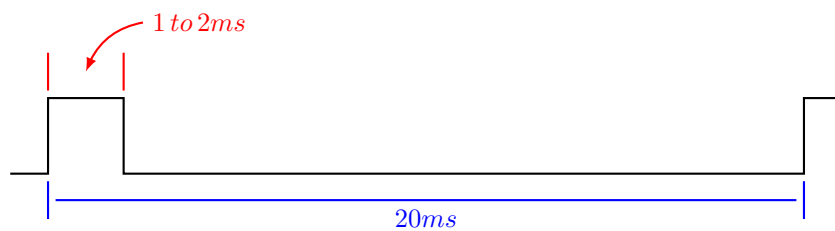


Figure 3: Timing Requirements for Servo PWM

2. Use your barrel jack adapter to supply 5V for the servo. Use your volt meter to confirm that the converter is providing 5V.
3. Having confirmed that the PWM signal and the supply voltage are correct, wire the servo motor onto your breadboard. Be sure to unplug your Nucleo from power while you make these changes. Verify that you can control the motor position safely by turning the shaft of the potentiometer. Once you've confirmed this, you can remove the pot from the setup.

Note!

There are several different color schemes for the servo motor connections. The center pin is VCC (5V), the darker of the outer wires (either black or brown) in GND, the other is the PWM signal. If you are unsure, ask an instructor or a TA.

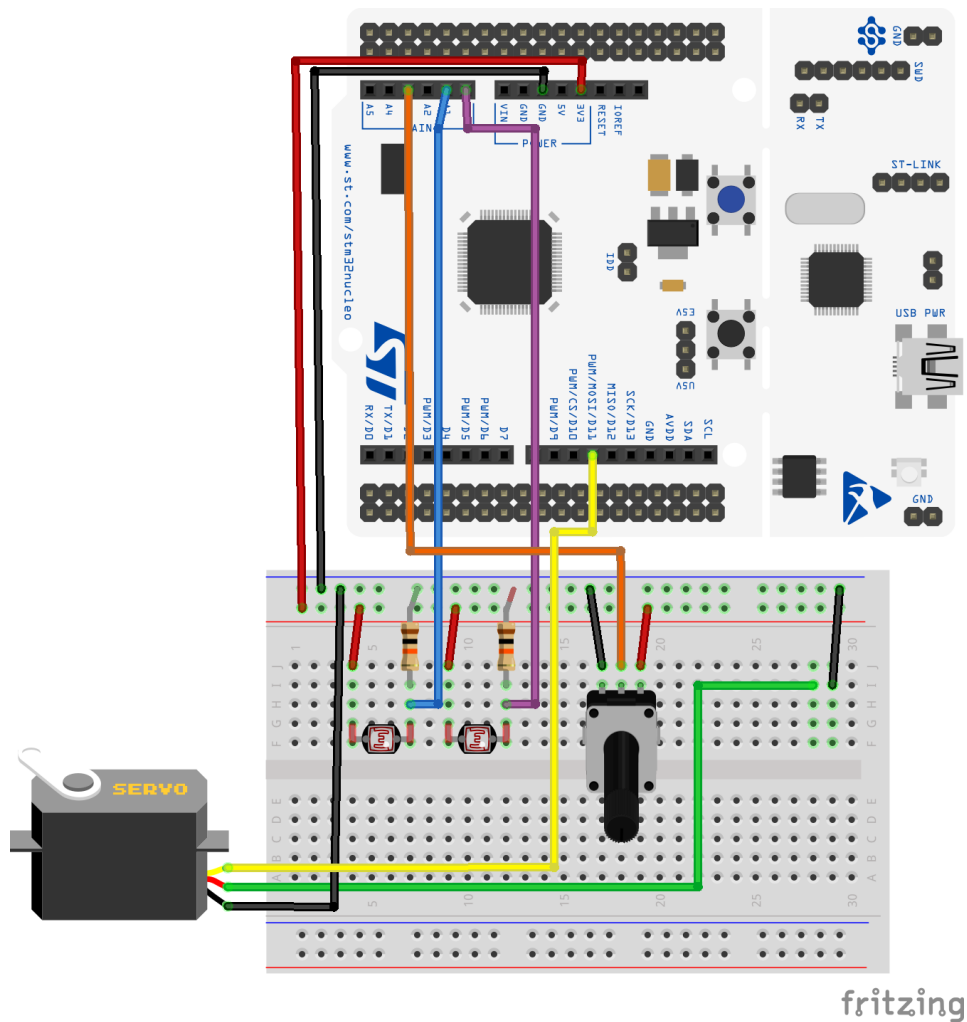


Figure 4: Servo Test Circuit

4 Design Challenge - Solar Tracker

A solar tracker uses servo motors to keep a solar panel pointed at the sun throughout the day. (You may have seen an example on campus between Kemeny and Moore; the photo in Figure 5 is not it).



Figure 5: A Typical Solare Tracker

Some trackers work by attaching photosensors at the edges of the solar panel, and using the sensor imbalance caused by oblique illumination to control servos in elevation as well as azimuth. We're going to build a model of an azimuth-only system depicted in Figure 6.

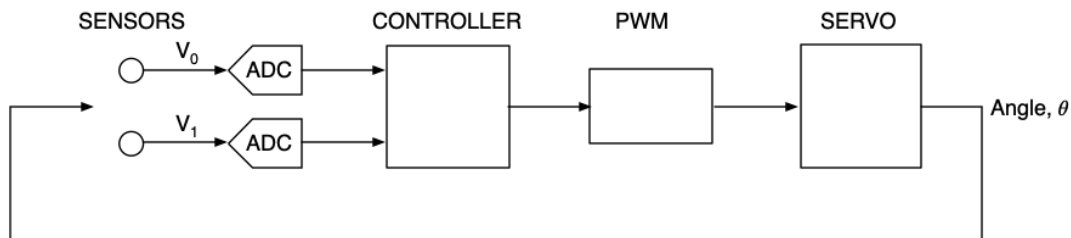


Figure 6: Azimuth Only Controller

Here is the theory of operation. The sensor voltages will differ according to the angle of the servo relative to the light source (your flashlight, simulating the sun). If $V_1 = V_0$, the tracker is pointing right at the source, as desired. If $V_1 \neq V_0$, you want to nudge the servo in a direction that will tend to equalize the voltages. The difference $V_1 - V_0$ is called the error, e . Repeated nudges, or steps, increase or decrease the PWM value, until the error is reduced to zero when the servo is pointing the sensors directly at the source.

The controller, which you will write in C, takes in the ADC values, updates the PWM value, and writes it to the CCR1 register. This is called a closed-loop feedback control system. You can learn more about them in ENGS 26.

The controller obeys a simple recursive equation:

$$PWM(t + T) = PWM(t) \pm \Delta$$

where t is the current time, T is the sampling interval (update time), and Δ is the step size, positive or negative as needed. A slightly fancier version uses a step size that is proportional to the error,

$$PWM(t + T) = PWM(t) + Ke(t)$$

where $e(t) = V_1(t) - V_0(t)$ is the error and K is the controller gain. This controller will take larger steps when the error is large, and smaller steps as the correct pointing is approached. In practice, we don't need to convert the ADC readings to actual voltages. We can just use the ADC readings directly and lump the proportionality constant $V_{REF}/4096$ into K .

You, as the designer, can adjust T and K to “tune” the response of the feedback loop. Part of the fun of this lab is experimenting with different values of these parameters. How smooth and responsive can you make the loop? It is possible, by being too aggressive, to get oscillatory behavior if the controller makes the servo shoot past its goal and have to backtrack. On the other hand, if you are too cautious, with too small a K or too large a T , the servo will always seem to be lagging behind the motion of the light source. I recommend that you start out cautiously, with a value of K on the order of 0.01, and increase it as you gain understanding of the system's dynamics.

Be sure that you impose upper and lower bounds on the PWM values before writing to OCR1A so that you don't overdrive the servo beyond $\pm 60^\circ$ of center.

You will be using Timer 14 for the PWM, so that you can achieve the necessary 50Hz PWM frequency. Use the millisecond timer we have provided keep track of your value for T .

When you test your system you'll want to make sure that your flashlight is brighter than any competing light source. My setup is attracted to my laptop screen, the lamp across the room, and even a sunny window. Write a complete program to implement the closed-loop servo control system. Keep a log of your tuning process, noting your observations as you adjust K and T .

Deliverable 9: Project Demo (10pts)

Demonstrate your working system to an instructor

Staff Signature _____

4.1 Final Touches Credit

Deliverable 10: Angle Display (2pts)

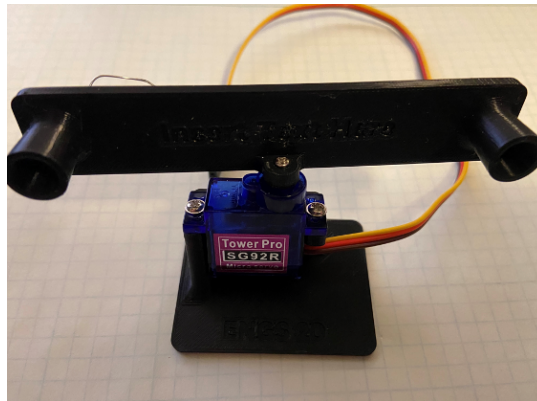
Put your seven-segment display back in the system, calculate and display the servo angle.

Staff Signature _____

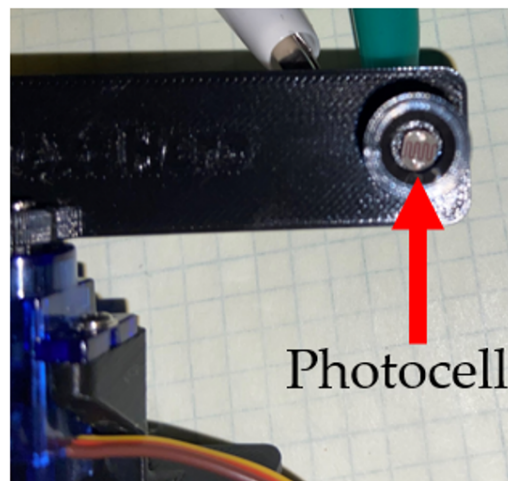
A Servo Assembly

Your kit includes a 3D-printed motor arm and stand for your solar tracker. Most have already been assembled. If yours has not, follow the directions below.

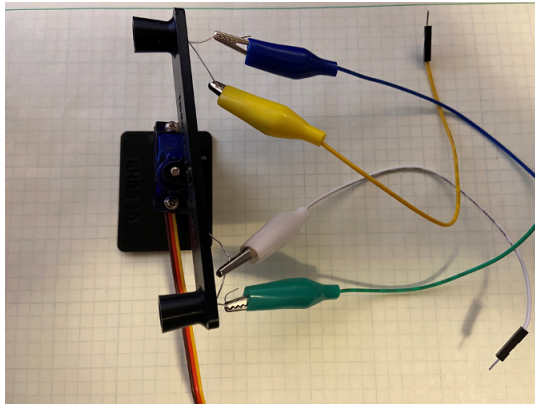
- Using the small screws in your servo kit, gently attach the to the mount, and then the arm to the motor. Be careful when screwing the motor arm into place—applying too much torque to the shaft could break the servo motor. Once assembled, your motor-mount-arm unit should look like this:



- Now insert your photocells into the shielding cups on the motor arm, like so:



- Use alligator-to-breadboard clips to make the electrical connections between the leads of the photosensors and your breadboard:



- Connect the leads of the alligator clips to your breadboard to recreate the photocell circuit from earlier in the lab.